t-tests:
Comparing two groups

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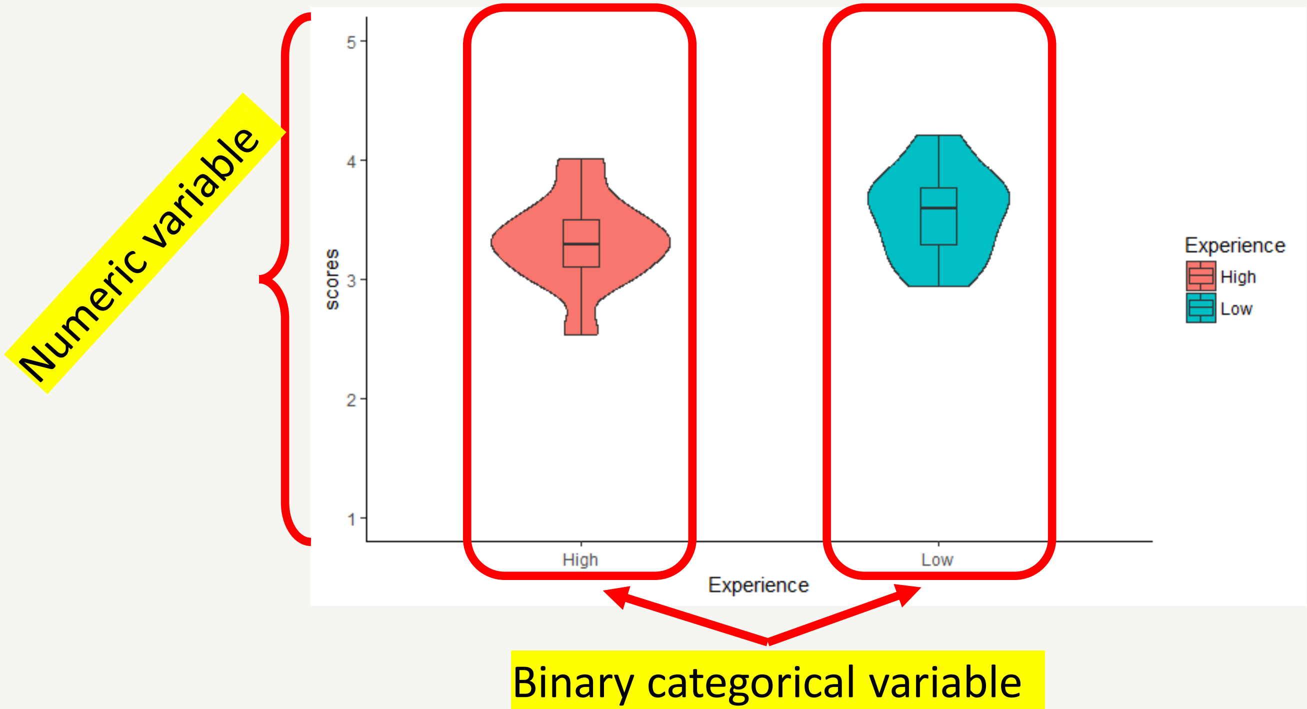
Learning goals

- Learn about the different **t-tests**
- Identify when you should use **which** t-test
- Understand how the **t-statistic** is calculated and what it means
- Know the **assumptions** of t-tests and how to test them
- **Conduct** t-tests in R and interpret output
- **Report** the results of t-tests
- Understand what an **effect size** is and why it is important
- Calculate **Cohen's D** for t-tests

When to use t-tests?

- T-tests are used when testing the relationship between a **binary independent variable** (two groups) and **a numeric dependent variable**



Remember the types of research design?

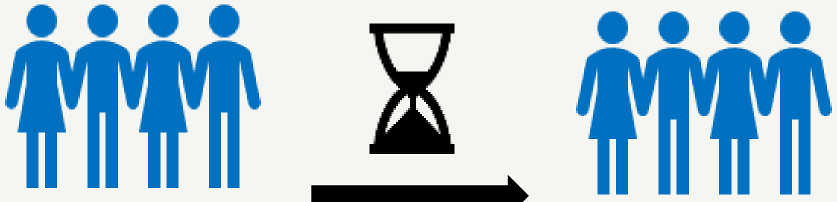
Between-subjects design

Comparing two mutually exclusive groups
a.k.a. *independent samples*

Group A   Group B

Within-subjects design

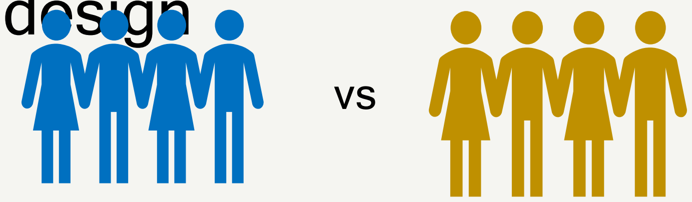
The same participants tested at different times
a.k.a. *repeated measures*



Time 1   Time 2

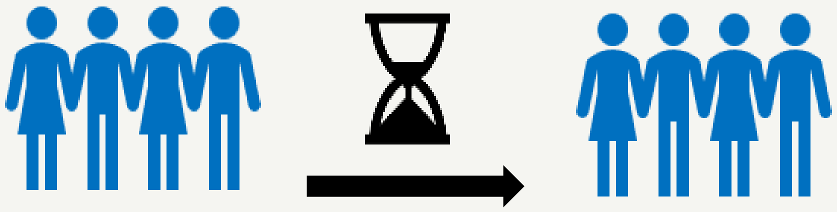
Types of t-test

Between-subjects design



Group A Group B

Within-subjects design



Time 1 Time 2

Two samples t-test

Do the means of two samples differ from each other?

e.g., *Do novice and experienced drivers differ in driving confidence?*

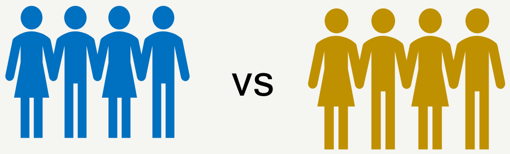
Paired samples t-test

Do the means of two ***paired*** samples differ from each other?

e.g., *Do people believe they are better drivers after they have been on a speed awareness course than before they went on the course?*

Types of t-test

Between-subjects design



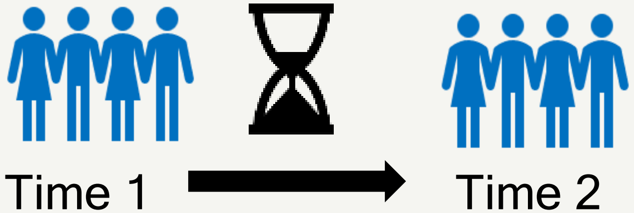
Group A Group B

Two samples t-test

Do the means of two samples differ from each other?

e.g., *Do novice and experienced drivers differ in driving confidence?*

Within-subjects design

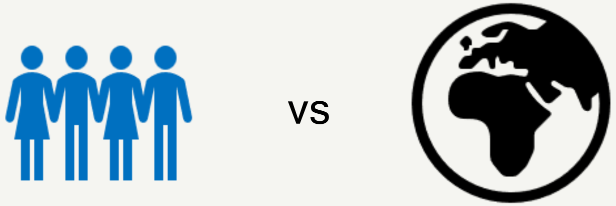


Time 1 Time 2

Paired samples t-test

Do the means of two *paired* samples differ from each other?

e.g., *Do people believe they are better drivers after they have been on a speed awareness course than before they went on the course?*



Group A Population mean

One sample t-test

Is a sample mean different to a known or specified population mean?

e.g., *Do people believe they are better than average drivers?*

One sample t-test

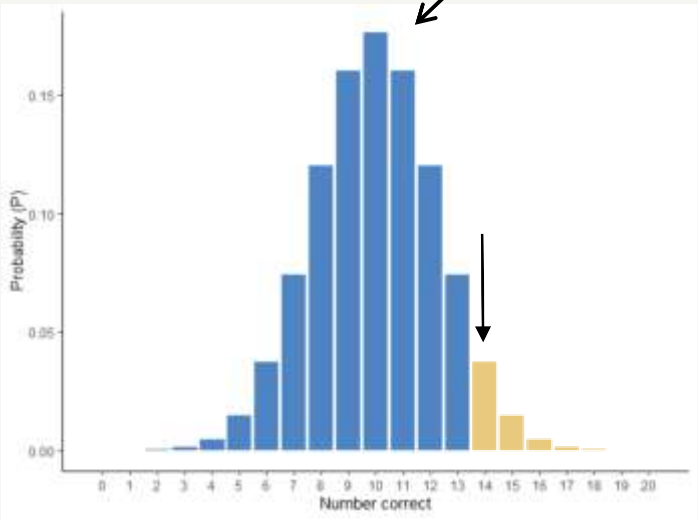
Example

Remember the steps of hypothesis testing

or

H_1 world
(real difference)

H_0 world
(no real difference)



- Step 1. Formulate the **null** and **alternative** hypotheses.
- Step 2. Choose a **test statistic** that describes the effect we are interested in (*t*)
- Step 3. Simulate how that statistic would be distributed **if H_0 is true**
- Step 4. Calculate the **probability (p)** that we might observe our test statistic if H_0 was true
- Step 5. Compare the probability to a **critical threshold** ($p<.05$) and retain/reject H_0 accordingly

One sample t-test

Do people believe they are better than average drivers?

We asked a sample of 25 drivers in the UK to rate how good they are at various aspects of driving on a 5-point scale with 10 items, e.g.:

How safe do you think your driving is?

1	2	3	4	5
Much worse than average	A bit worse than average	Average	A bit better than average	Much better than average

Each participant then gets a mean score on perceived driving skill between 1-5

Hypotheses

- If people were *accurate* at assessing their driving skill, we would expect the population mean to be 3 on this questionnaire (indicating average driving).
- Our **alternative hypothesis** is the people will report being better drivers than average. E.g., the true population mean will be greater than the hypothesised population mean of 3.
- Our **null hypothesis** is that our true population mean is not different to our hypothesised population mean of 3.

H_1 = population mean > 3

H_0 = population mean ≤ 3

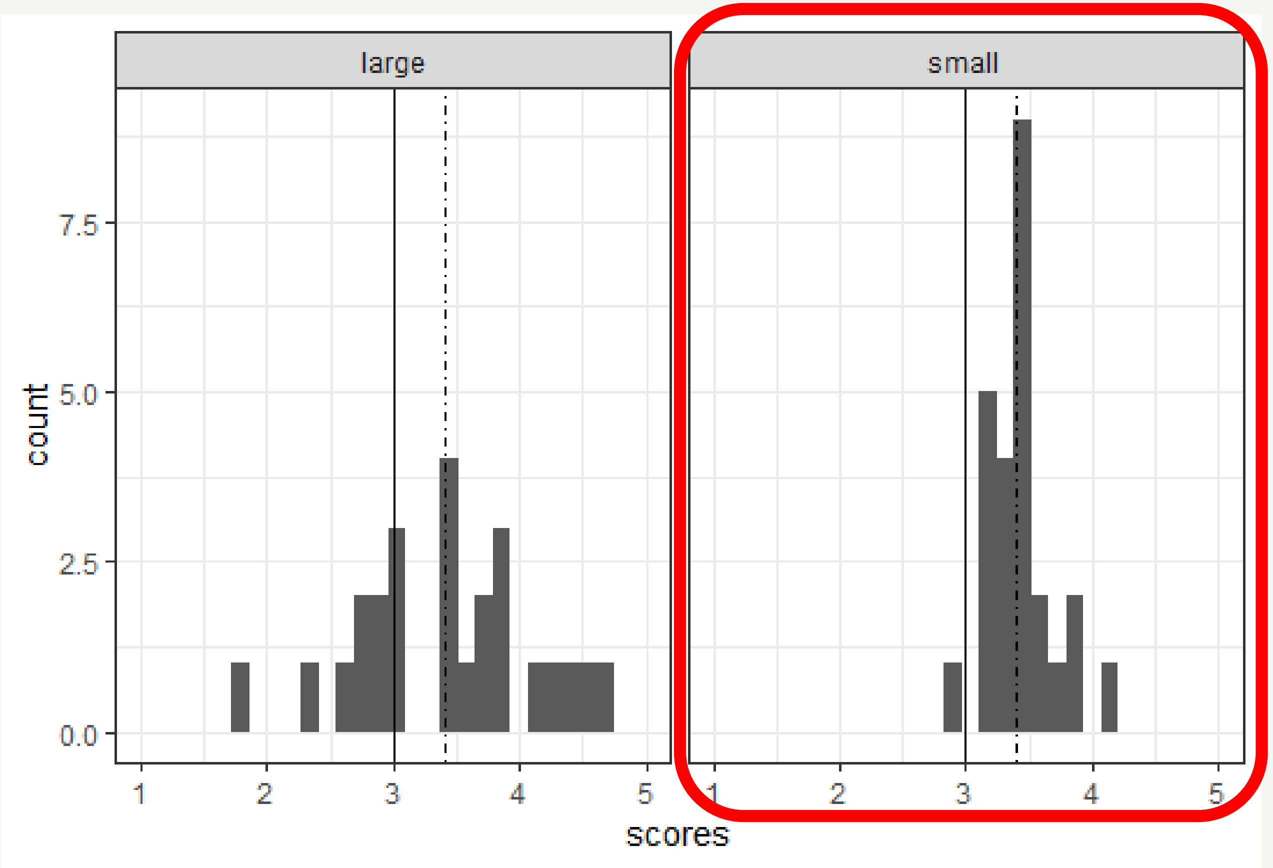
Note this is a directional hypothesis!

Test statistic

- What should our test statistic be?
- Our sample mean = **3.4**
- Expected mean = **3**
- So, the mean difference is **0.4**
- Is a difference of 0.4 enough information to determine the likelihood that our sample mean came from a population with a mean of 3?

  No - this difference might be entirely due to **sampling error**
- We also need to take the variation in the data into account…

Variation matters



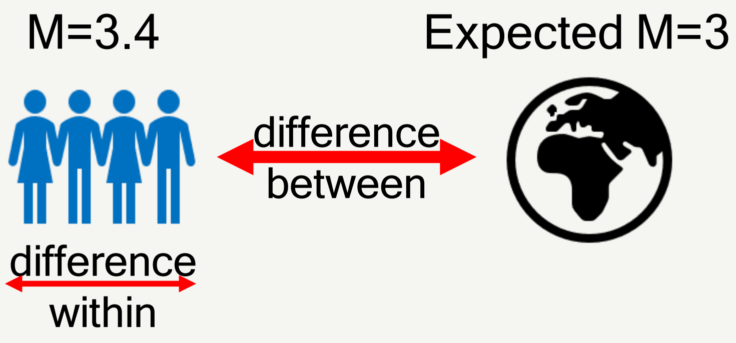
Both these samples have M=3.4, but…

this one has much ***less variance***

➔ much more convincingly different from 3.

T-statistic

- To get a measure of **confidence** in our effect, we need to take into account both:
  - **the effect** (mean difference)
  - **the error** (variation)



T-statistic

- To get a measure of **confidence** in our effect, we need to take into account both:
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- To do this, divide our effect by a measure of error

$$t = \frac{effect}{error}$$

T-statistic

- To get a measure of **confidence** in our effect, we need to take into account both:
  - **the effect** (mean difference)
  - **the error** (variation)
- To do this, divide our effect by a measure of error
- From the lecture on sampling: what do we use as a measure of error when sampling from a population?

$$t = \frac{effect}{error}$$

Standard Error (SE) of the Mean

=

$$\frac{SD\ (sample\ standard\ deviation)}{\sqrt{sample\ size}}$$

T-statistic

- The t-statistic is a measure of the difference between means *taking into account* the variability in the data.

$$t = \frac{effect}{error} = \frac{difference\ between\ means}{Standard\ Error}$$

- Larger difference between means ➔ larger *t*
  - (higher confidence in our mean difference)
- Larger variability ➔ smaller *t*
  - (lower confidence in our mean difference)

Calculating one sample T value

General t formula

$$t = \frac{difference\ between\ means}{Standard\ Error}$$

One sample t formula

$$t = \frac{sample\ mean\ -\ hypothesised\ population\ mean}{SD/\sqrt{sample\ size}}$$

$$t = \frac{3.423-3}{0.354/\sqrt{25}} = 5.97$$

Remember the steps of hypothesis testing!

- Step 1. Formulate the **null** and **alternative** hypotheses.
- Step 2. Choose a **test statistic** that describes the effect we are interested in (t)
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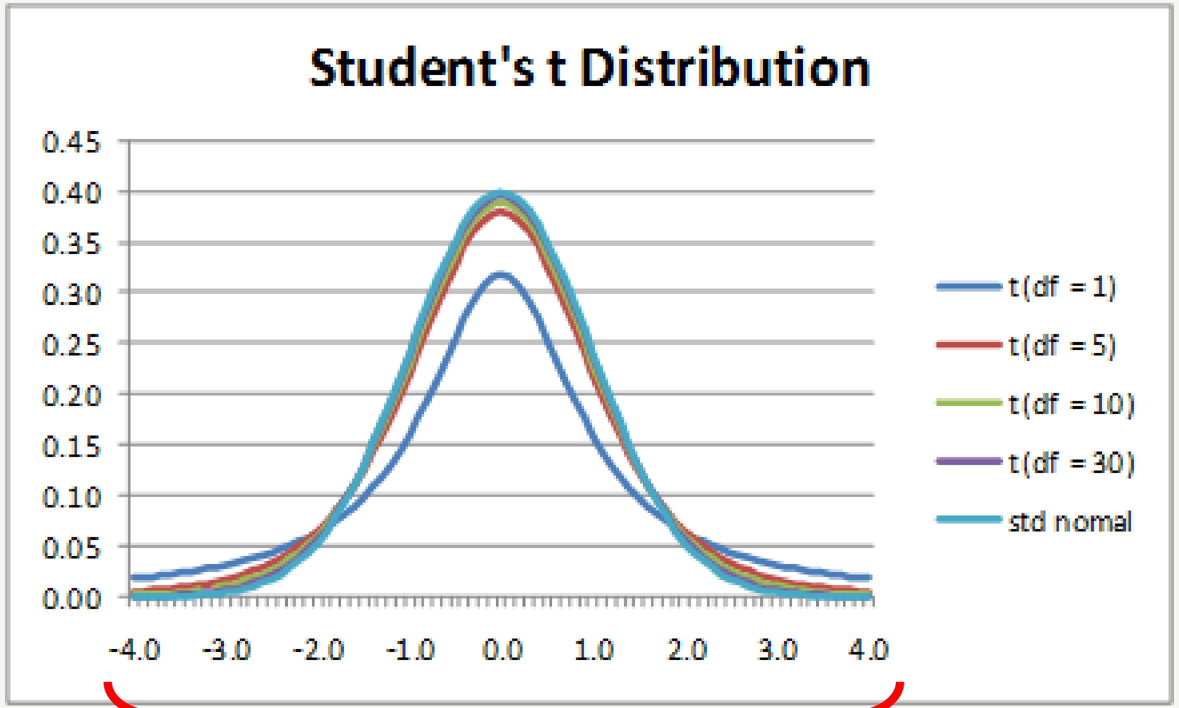
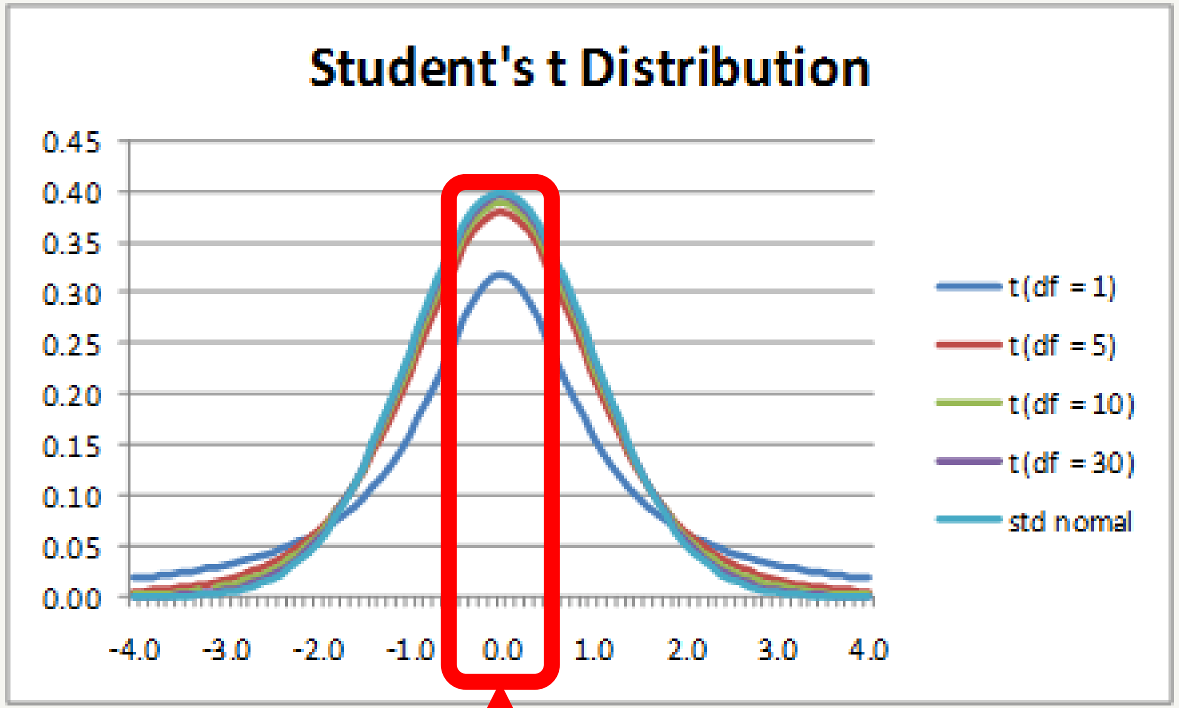
Null hypothesis distribution



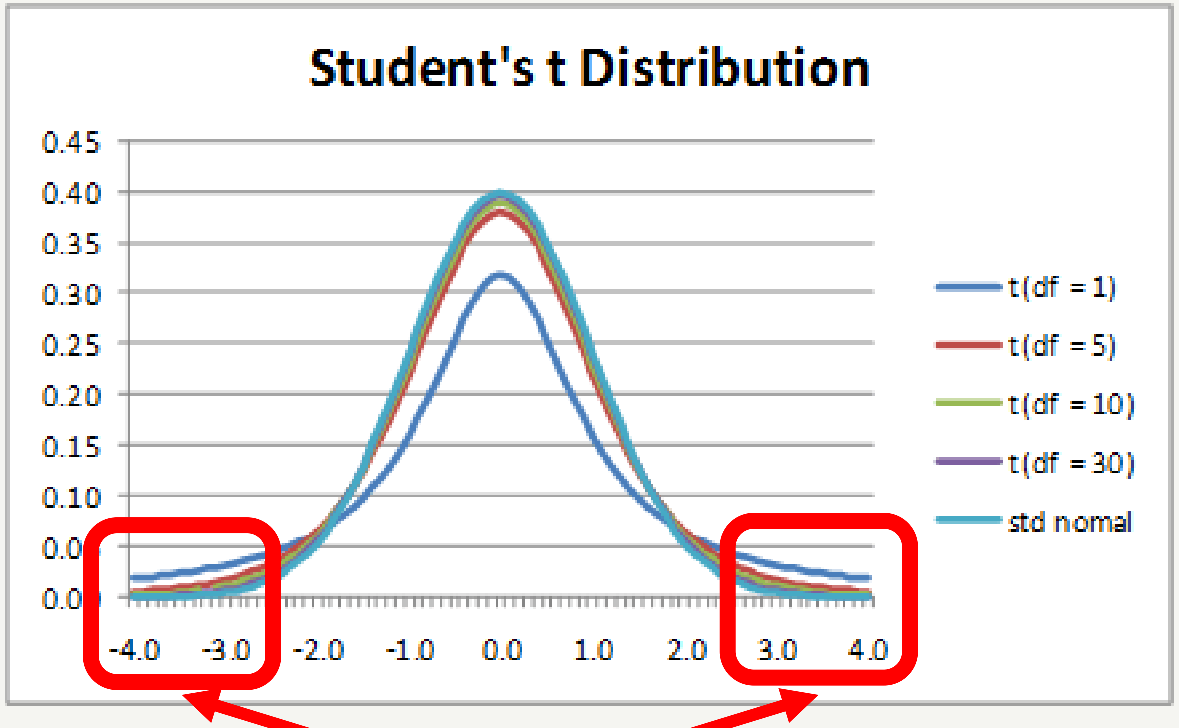
This is a range of possible *t*-values assuming a null hypothesis (no difference)

Null hypothesis distribution



If H_0 is true, you'll find no difference (t=0) or a very small difference most of the time

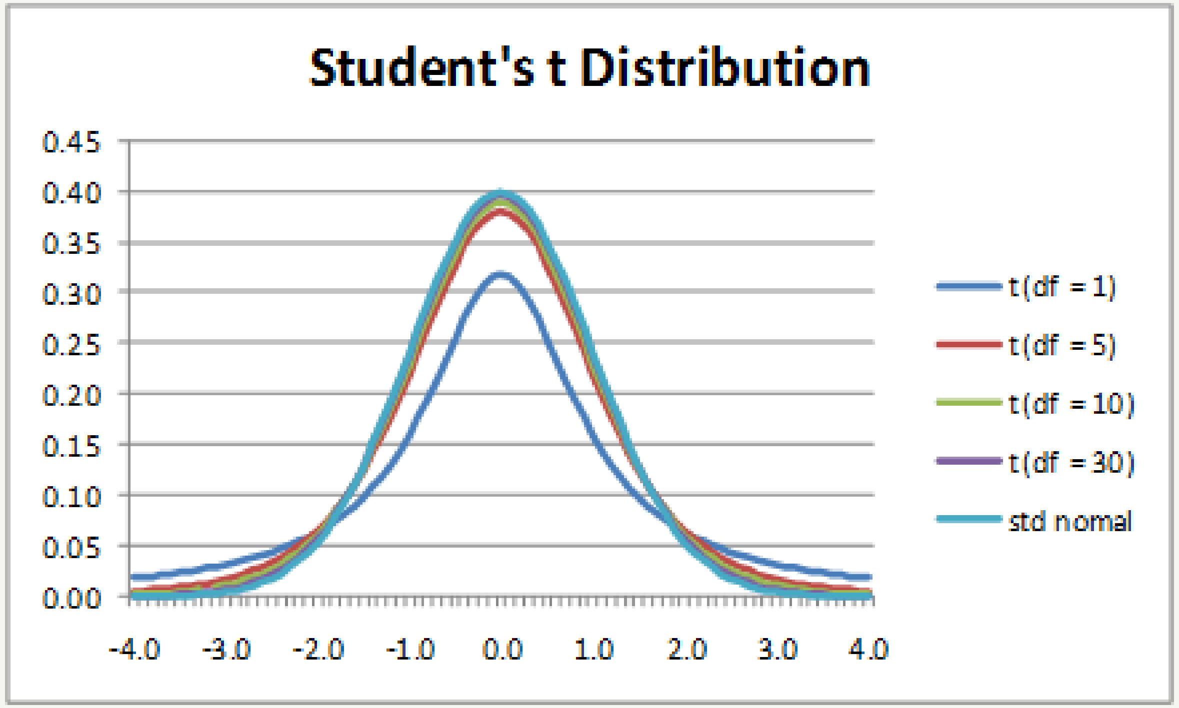
Null hypothesis distribution



If H_0 is true, you'll find large differences only very rarely

Null hypothesis distribution

- For large samples, the distribution of H_0 t-values has the standard normal distribution.
- For smaller samples (fewer degrees of freedom) the t-distribution has fatter tails
- This is because we don't know the true population standard deviation and our sample SD underestimates the population SD



Remember the steps of hypothesis testing!

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Look up P-value

- We can do this in R using the t distribution probability function `pt()`.
  - More traditionally, you could look it up in a book.
- For `pt()` we need to enter:
  - the t value,
  - the degrees of freedom
  - `lower.tail = FALSE` (to look at the portion of the distribution *above* the sample t value)

```
➤pt(5.97, df=24, lower.tail = FALSE)
➤[1] 1.834166e-06
```

Look up P-value

```
>pt(5.97, df=24, lower.tail = FALSE)
>[1] 1.834166e-06
```

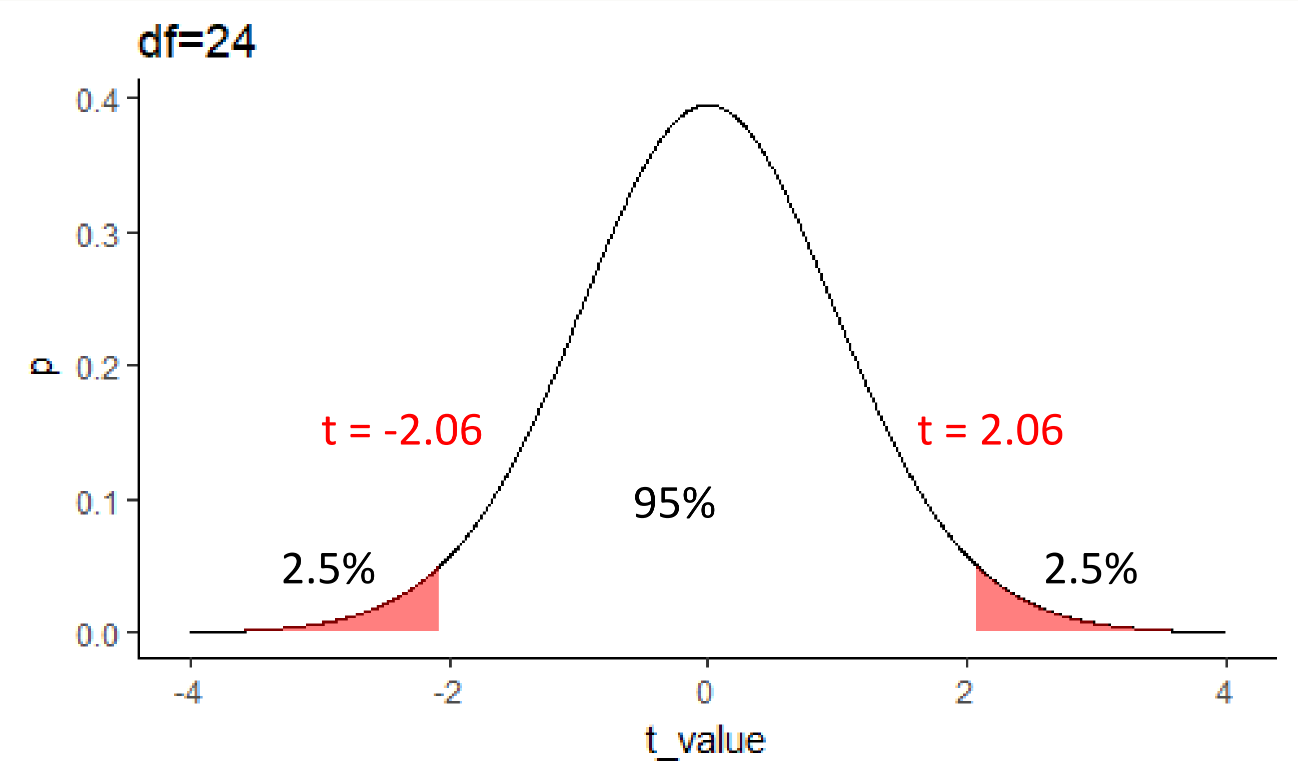
- This is the p-value which is presented here in scientific notation. P is a really, really small number. In this case we usually write $p < .001$.
  - This means that there is a very small probability (< .001) that we would have observed our data if the null hypothesis was true.
  - For a two-sided hypothesis test (which is typical) we multiply $p \times 2$ to get probability of a value this extreme on *either* side of the null distribution.

```
> 1.834166e-06*2
[1] 3.668332e-06
```

Alpha

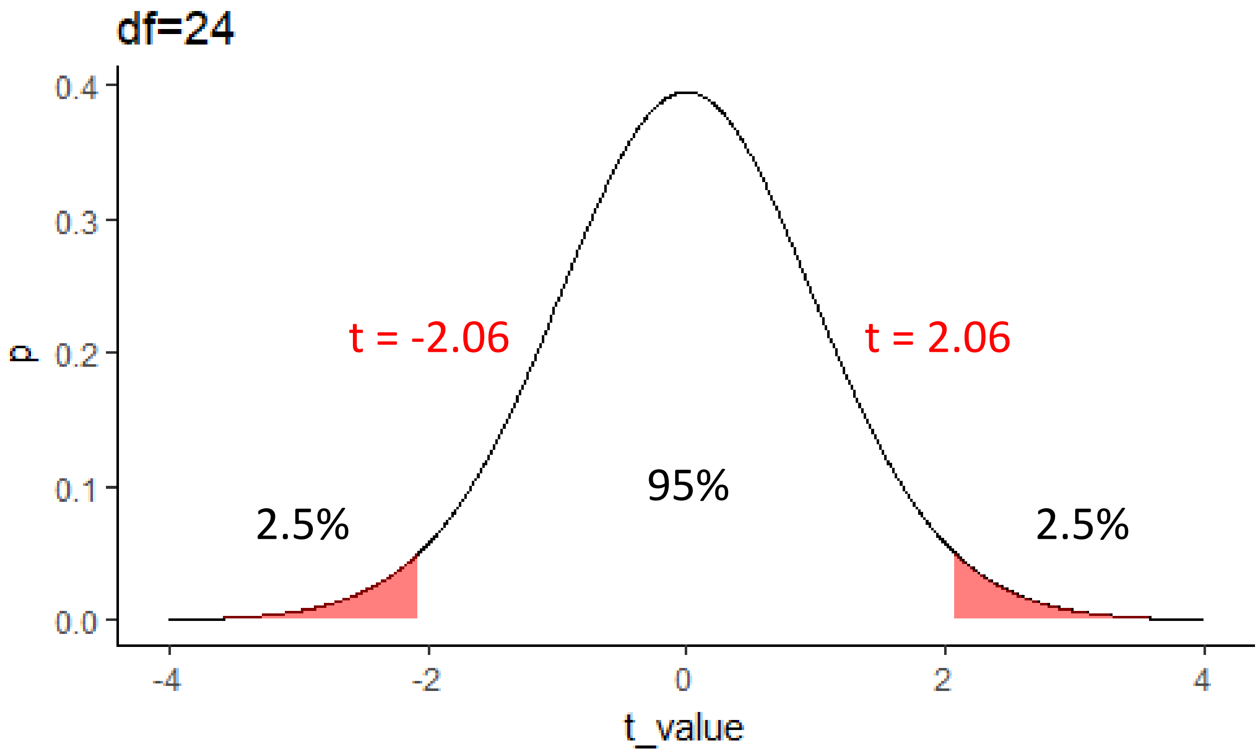
For the t-distribution with 24 degrees of freedom and an alpha level of p = .05 the critical t-value is 2.06.

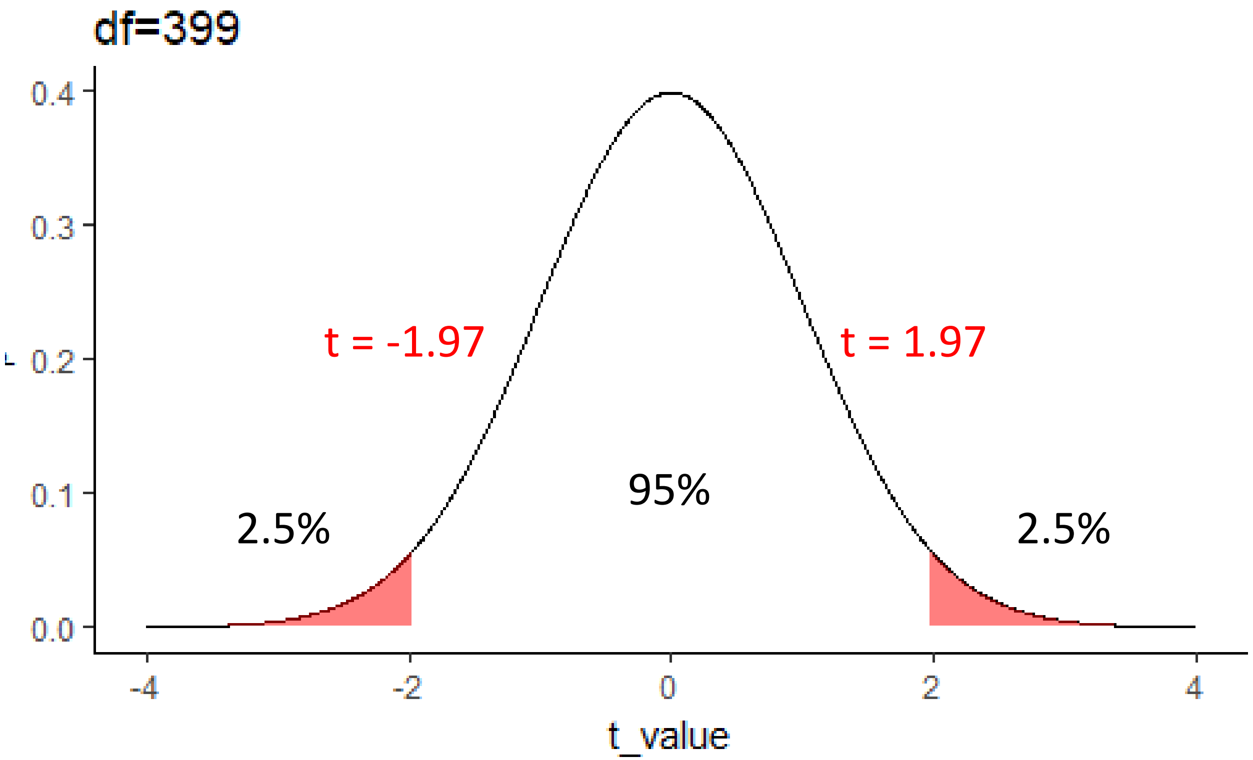
t-values above 2.06 or below -2.06 are considered significant.



Critical t-value

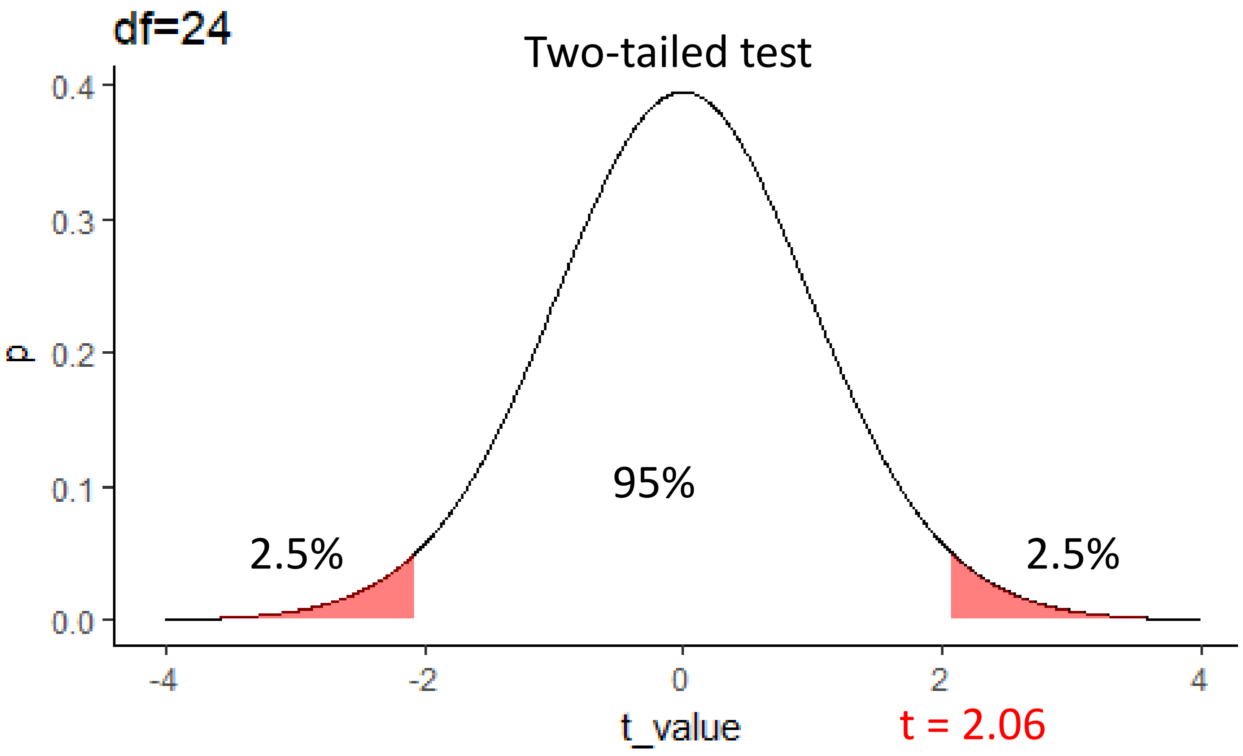
As sample size (degrees of freedom) increases, critical *t*-value becomes smaller

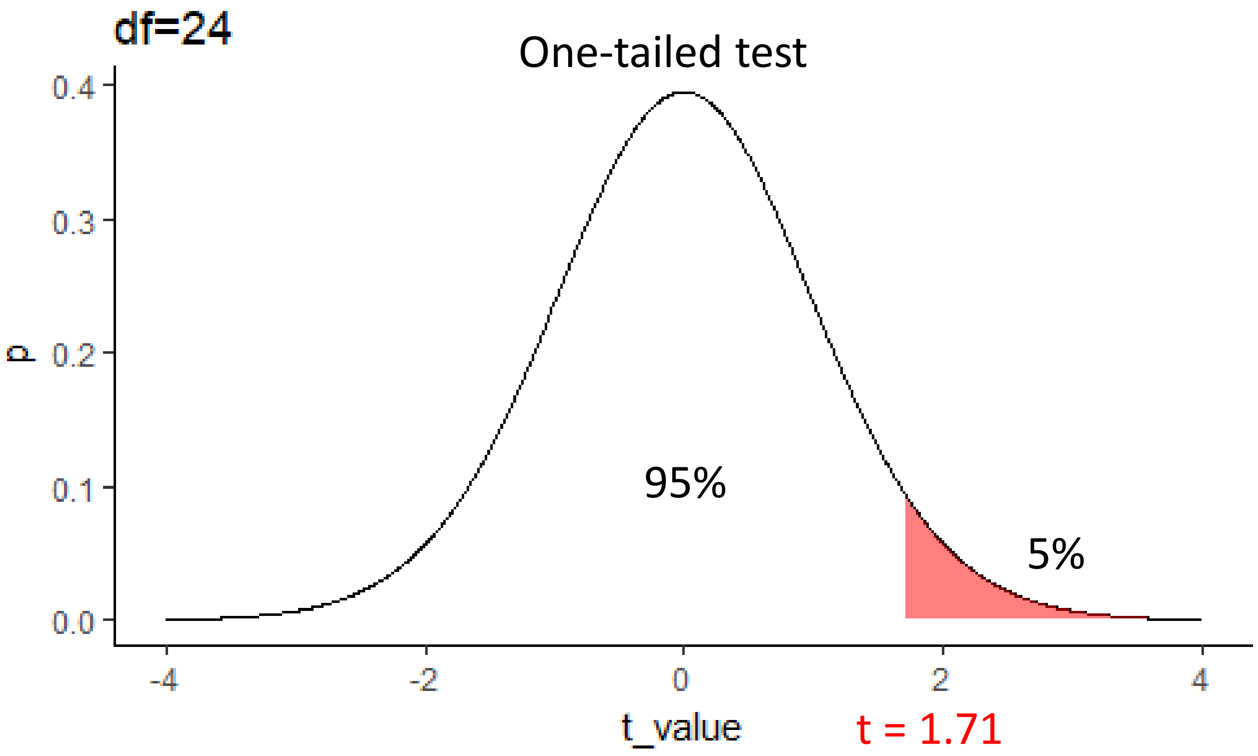




One tailed vs two tailed tests

Critical t-value also depends on whether it is a *two*-tailed of *one*-tailed test





Compare p-value to Alpha

- The critical alpha value is typically p = .05
  - ➔ we will reject H_0 if p < .05 (which it clearly is in this case)
- Another way of thinking about p-values: they indicate the risk of Type 1 error (false positive)
  - If the p-value is .049 then we have to accept that in 1 out of 20 experiments we will reject H_0 when we shouldn't
- In our case (p< .001), we would be making a Type 1 error in fewer than 1 in 1000 studies.

One sample t-tests in R

```
t.test(data %>% pull(scores), mu = 3, alternative = "greater")
```

Pulls the scores of our sample

The population mean (mu) to compare the sample against

The alternative hypothesis is that the true population mean is greater than 3

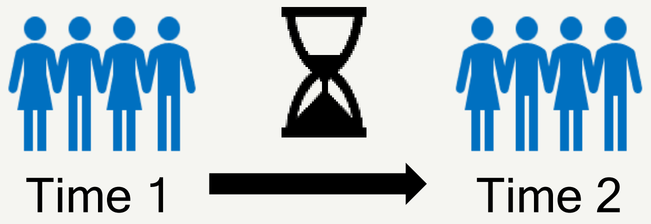
One sample t-tests in R

```
t.test(data %>% pull(scores), mu = 3, alternative = "greater")
```

```
[output]
One Sample t-test data: data %>% pull(scores)
t = 5.9681, df = 24, p-value = 3.685e-06
alternative hypothesis: true mean is not equal to 3
95 percent confidence interval:
3.276731 3.569308
sample estimates:
mean of x
3.423019
```

Paired t-test

Example



Paired t-test example

Does a driving intervention increase or decrease drivers' confidence in their driving skill?

We ask a sample of 25 drivers to rate how good they are at various aspects of driving on a 5 point scale with 10 items before and then after intervention.

How safe do you think your driving is?

1	2	3	4	5
Much worse than average	A bit worse than average	Average	A bit better than average	Much better than average

Each participant has a pre-intervention score and a post-intervention score for perceived driving skill between 1-5

Hypotheses

- If the intervention didn't change driving confidence, we would expect the population mean difference to be 0.
- Our **alternative hypothesis** is that people's perception of their driving skill will be different after the intervention. E.g. the true population mean difference is not 0.
- Our **null hypothesis** is that people's perception of their driving skill will be the same before and after the intervention. Population mean difference is 0.

H_1: population mean difference $\neq 0$

H_0: population mean difference $= 0$

Paired sample T value

General t formula $$t = \frac{difference\ between\ means}{Standard\ Error}$$

One sample t formula $$t = \frac{sample\ mean - hypothesised\ population\ mean}{Sample\ standard\ deviation/\sqrt{sample\ size}}$$

Paired sample t formula $$t = \frac{sample\ mean\ difference - \mathbf{0}}{sample\ standard\ deviation\ difference/\sqrt{sample\ size}}$$

Paired sample t test is a one sample t test on the difference with hypothesised population mean difference of 0.

Running the test in R: data set up

Measurement 1

Measurement 2

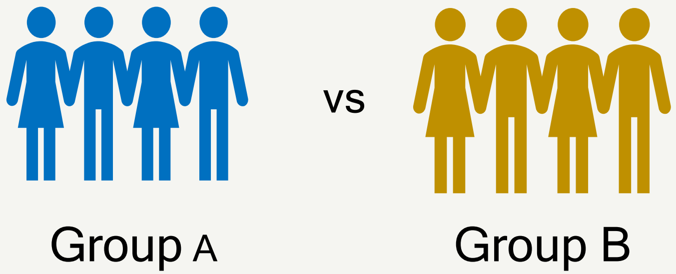
Filter

	id	before	after
1	1	4.067029	3.528748
2	2	3.354520	3.582129
3	3	4.755535	5.321291
4	4	2.805218	2.875698
5	5	3.732931	4.289700
6	6	4.205236	4.608816
7	7	2.871051	2.758506
8	8	4.142891	4.639057
9	9	3.591477	3.807124

Is this long or wide format?

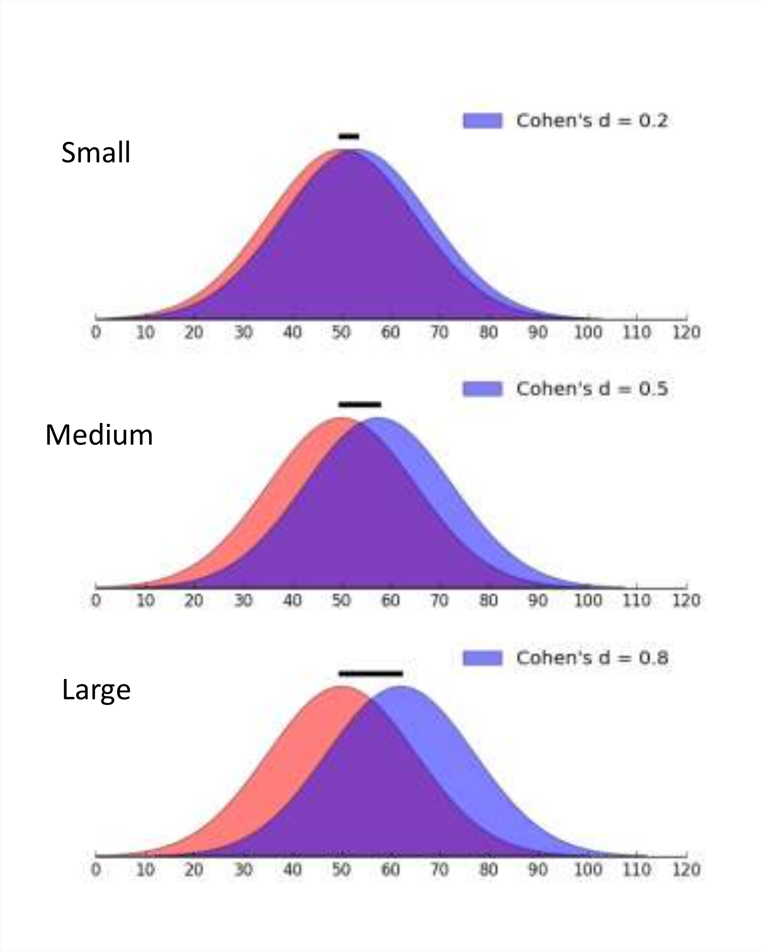
Running a paired t-test in R

```
```{r}
t.test(dat %>% pull(after),dat %>% pull(before),paired = TRUE)
```
```

```
        Paired t-test

data:  dat %>% pull(after) and dat %>% pull(before)
t = 3.5974, df = 24, p-value = 0.001447
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.1038686 0.3834573
sample estimates:
mean of the differences
              0.2436629
```

Two-samples t-test

Example



Two-samples t-test

- **Is there a difference in confidence between novice and experienced drivers?**
- In a new study we ask 25 newly qualified and 25 experienced drivers to rate how good they are at various aspects of driving on a 5 point scale with 10 items, e.g.:

How safe do you think your driving is?

1	2	3	4	5
Much worse than average	A bit worse than average	Average	A bit better than average	Much better than average

- Each participant then gets a mean score on perceived driving skill between 1-5
- We calculate a mean for the novice group and a mean for the experienced group

Hypotheses

- Our **alternative hypothesis** is that there will be a difference in driving confidence between experienced and novice drivers.
  - E.g., *the population means will be different.*
- Our **null hypothesis** is that experienced and novice drivers are equally confident in their driving.
  - E.g., *the population means will not differ.*

H_1: experienced population mean ≠ novice population mean

H_0: experienced population mean = novice population mean

Note this is a non-directional hypothesis!

Data set up

Grouping (independent) variable

Dependent variable

ID	Experience	Scores
1	High	3.38
2	Low	2.70
3	Low	3.86
4	High	3.19
5	Low	2.97
…	…	…

Two sample t-test in R

```
> t.test(data %>% filter(Experience=="Low") %>% pull(scores),
      data %>% filter(Experience=="High") %>% pull(scores))

Welch Two Sample t-test
data: data %>% filter(Experience == "Low") %>% pull(scores) and data %>%
filter(Experience == "High") %>% pull(scores)
t = 2.3185, df = 47.996, p-value = 0.02473
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
0.03146643 0.44248971
sample estimates:
mean of x mean of y
3.554546 3.317568
```

Effect size

- t-statistics and p-values are dependent on the size of the sample.
- The t-value will increase and the p-value will decrease as the sample size increases, even if the size of the difference and variation in the data remains the same.
- When samples are large, trivially small differences may still be significant
- Statistically significant but unimportant? To clarify findings, we need to estimate the **effect size.**

Cohen’s D - a measure of effect size

- $Cohen's\ d = \dfrac{Mean\ difference}{Standard\ deviation}$

One-sample t-test:

```
cohensD(data %>% pull(scores), mu=3)
```

Two-samples t-test:

```
cohensD(data %>% filter(experience == “Low”) %>% pull(scores),
        data %>% filter(experience == “High”) %>% pull(scores),
        method = “unequal”) # in cases where the groups’ sample
                            # sizes and variances are unequal
```

Paired samples t-test

```
cohensD(data %>% pull(before),
        data %>% pull(after),
        method = “paired”)
```

```
```{r}
cohensD(dat %>% pull(before),dat %>% pull(after),
    method="paired")
```

[1] 0.7194792
```



Reporting t-tests

Reporting one sample t-tests

*Mean driving confidence scores were 3.43, 95% CI[3.28, 3.57], which is significantly **higher** than the expected mean score of 3 if drivers believed they were average; t(24) = 5.97, p < .001* (one-tailed)*; representing a large-sized effect (d = 1.21).*

Direction

t-value and degrees of freedom

p value < .001

effect size and Cohen’s d

Reporting paired samples t-tests

*Driving confidence scores were **higher** after the intervention (M = 3.90, SD = 0.87) than before the intervention (M = 3.66, SD = 0.69). This increase of 0.24, 95% CI[0.10, 0.38] was statistically **significant**; t(24) = 3.60, p = .001; and represented a medium-sized effect (d=.72).*

- Direction
- Descriptives
- Significance
- t-value and degrees of freedom
- Exact p-value
- Effect size and Cohen's d

Reporting a two-sample t-test

*Novice drivers reported **higher** scores on the driving confidence test than experienced drivers (novice M = 3.55, SD = 0.36, experienced M = 3.32, sd = 0.36). The mean difference of 0.22, 95% CI [0.031 0.442] was statistically significant; t(48) = 2.32, p = .03; and represented a medium-sized effect (d=.61).*

- Direction
- Descriptives
- Significance
- t-value and degrees of freedom
- Exact p-value
- Effect size and Cohen's d

Assumptions of a t-test

t-test assumptions

- DV is interval or ratio level (Likert scales are quasi-interval)
- DV is normally distributed (in each group if two sample test, the difference if paired sample test)
- For *two-samples* test: the two samples are independent
  - (in psychology this usually means different people take part in each condition)
- For *paired-samples* test: the two samples are paired

Student's vs Welch's

- Note that there are two different two sample t-tests
  - **Students t-test** assumes equal variances (assumes the standard deviations of the two populations will be equal even if means differ)
  - **Welch's t-test** (the one we have discussed) does not assume equal variances

Always use this one – it is the default in R

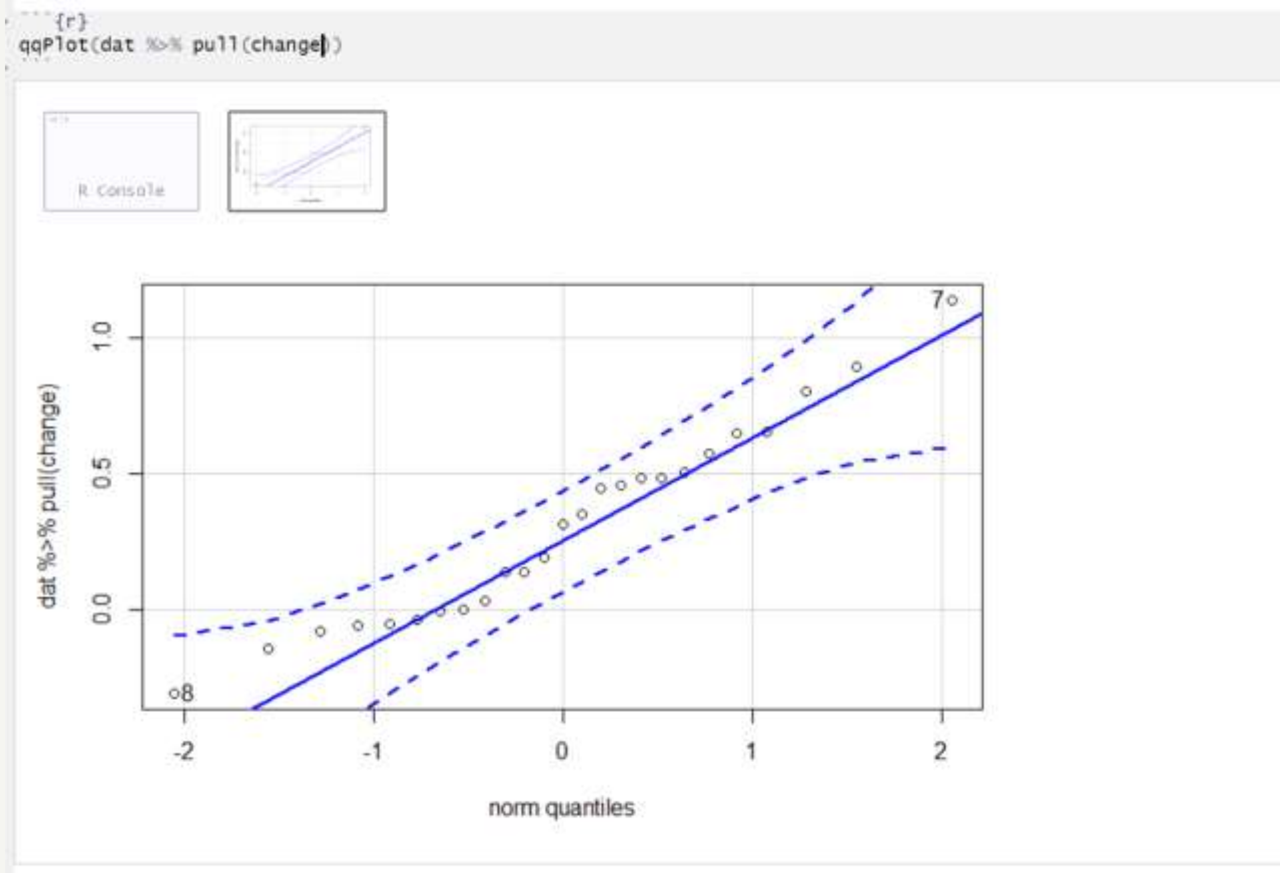
Testing assumption of normality

Quantile-Quantile plot (Q-Q plot)

From the package `car`

Q-Q plots show correlation between theoretical normal distribution quantiles and quantiles in the data

Normal data will lie along the straight line or at least within the dashed lines (confidence intervals)



Testing assumption of normality

Shapiro-Wilk test

Shapiro-Wilk compares the distribution of the data to a normal distribution. Its *null* hypothesis is that the data are normally distributed.

➔ A significant result ($p < .05$) suggests that the data differ from the normal distribution.

Use in combination with visualisation.

```{r}
shapiro.test(dat %>% pull(change))
```

```
	Shapiro-Wilk normality test

data:  dat %>% pull(change)
W = 0.96192, p-value = 0.454
```

Summary

- **One-sample** t-tests test the hypothesis that a population mean is different from a known or theoretical population mean.
- **Two-sample** t-tests test the hypothesis that two samples come from two different underlying populations.
- **Paired sample** t tests are used when you have a within-subjects dichotomous predictor.
- *All* require an outcome variable that is interval or ratio level
- T-tests are run using the `t.test()` function in R
- Assumption across all t-tests: that the outcome variable is normally distributed. You can test this assumption in R using histograms, Q-Q plots and the Shapiro Wilk test.
- Measures of effect size are helpful for interpreting your results in comparison to other studies. For t-tests this measure is Cohen's D.

Thank you